

		Mark	Remarks
Task 1.1	Defines the function encrypt(plaintext)	1	1
	Converts each character of the plaintext into its ASCII value (ord())	1	1
	Applies the correct increment to each character (message length decreasing by 1)	1	1
	Correctly wraps values exceeding ASCII 126 back to ASCII 32	1	1
	Converts the modified ASCII values back into characters (chr())	1	1
	Executes the specified test statement and obtains the expected output (True)	1	1
	Sub-total	6	6
Task 1.2	Defines the function decrypt(ciphertext)	1	1
	Converts each encrypted character into its ASCII value	1	1
	Correctly reverses the encryption by subtracting the appropriate values	1	1
	Correctly handles wrap-around from ASCII 32 back to ASCII 126 when required	1	1
	Executes the specified test statement and obtains the expected output (True)	1	1
	Sub-total	5	5
Task 1.3	Copies the encrypt(plaintext) function into the client program	1	1
	Copies the decrypt(ciphertext) function into the server program	1	1
	Creates, binds and listens on server socket	1	0
	Accepts client connection	1	0
	Receives encrypted message	1	0
	Decrypts received message	1	1
	Displays encrypted message	1	1
	Displays decrypted message	1	1
	Opens LOG.csv in append mode and writes a log entry	1	0
	Log entry contains Date/Time, IP, Port, encrypted and decrypted messages	1	0
	Prompts whether to continue listening	1	1
	Continues listening when user enters Y	1	1
	Terminates server correctly when user enters a value other than Y	1	1
	Sub-total	13	8
Task 2.1	Reads data from Teachers.csv	1	1
	Creates a MongoDB client connection and accesses school database and teacher collection	1	1
	Converts data to the appropriate data types before insertion	1	0
	Does not insert fields with empty values into the document	1	1
	Inserts all teacher documents into the teacher collection	1	0
			insertion is never reset for each row
	Sub-total	5	3
Task 2.2	Queries all documents from the teacher collection	1	0
	Displays all required fields for every document retrieved	1	1
	Displays '-' for fields that do not exist in a document	1	0
	Sub-total	3	1
Task 2.3	Queries the teacher collection and displays only the _id and Name fields	1	0
	Correctly identifies teachers whose documents do not contain the Contact field or the Role field	1	0
	Correctly identifies teachers whose Rating value is below 1600	1	0
	Sub-total	3	0
Task 2.4	Identifies one teacher from the Physics department	1	
	Updates the teacher's department to Computing	1	
	Sub-total	2	0
Task 3.1	Creates a Flask web application	1	1
	Routes to and displays the Menu.html web page	1	1
	Provides navigation to the Delete Erroneous Certs page	1	1
	Provides navigation to the VIP Voucher Recipients and Add Certificate Record pages	1	1
	Saves the displayed web page as required Task3_1	1	1
	Sub-total	5	5
Task 3.2	Creates a Flask route for the Delete Erroneous Certs page	1	1
	Uses an SQL DELETE statement to remove records with Status = 404	1	1
	Executes the SQL statement from the Flask application	1	1
	Displays the number of deleted records clearly on a web page	1	0
	Saves the displayed web page as required Task3_2	1	0
	Sub-total	5	3
Task 3.3	Creates a Flask route for the VIP Voucher Recipients page	1	1
	SELECT query retrieves all required fields	1	1
	Correct joins between the three tables	1	1
	WHERE clause retrieves certificates issued after January 2023	1	1
	ORDER BY LastName, FirstName ASC	1	0
	Executes the query	1	1
	Creates an HTML web page and displays the query results in an HTML table	1	1
	Saves the displayed web page as required Task3_3	1	0
	Sub-total	8	6
Task 3.4	Creates a Flask route for the Add Certificate Record page	1	
	Creates and displays an HTML web form	1	
	Accepts input for MemberId, CourseCode and IssueDate	1	
	Automatically sets Status to 200	1	
	Inserts a new record into the Certificate table	1	
	Uses the values entered in the web form in the SQL INSERT statement	1	
	Displays the entered details after submission	1	
	Displays a success message when the record is inserted successfully	1	
	Displays an error message when the record cannot be inserted	1	
	Successfully processes the first given test record and displays the required output	1	
	Successfully processes the second given test record and displays the required output	1	
Coding Style	Meaningful identifiers	1	1
	Whitespaces	1	1
	Comments	1	1
		1	0
	Sub-total	4	3
TOTAL		70	40